



Pilibrico Company LLC
1010 N. Hooker Street
Chicago, IL 60642
Ph. 312 337-9900 Fax 312 337-9903
www.Pilibrico.com

Technical Data Sheet

Plicast HyMOR 3100 Special KK

Product Number 2428

Date 1/1/2013

Product Description

A versatile high strength, mullite based low cement castable with excellent resistance to thermal shock/cycling. May be vibration cast, cast or pumped.

Service Limit - 3100°F 1,704. °C

Density to place 152 pcf 2,435 kg/m³

Density in service 152 pcf 2,435 kg/m³

Min Time before firing 16 hr

Std. package 55 # / 25 kg Bag

Water range per std. package:

Casting 1.6 to 1.8 qts 1.6 to 1.7 l

Pumping 1.9 to 2.0 qts 1.8 to 1.9 l

Chemistry % (calcined)

Al ₂ O ₃	61.1	P ₂ O ₅	
SiO ₂	34.7	Alk.	0.2
Fe ₂ O ₃	0.9	MgO	0.1
CaO	1.6	SiC	
TiO ₂	1.6	ZrO ₂	
Other			

Thermal Conductivity

	btu*in/hr*in ² * F ⁻¹	w/m°C
500F / 260C	10.7	1.54
1000F / 540C	11.3	1.62
1500F / 815C	12.0	1.72
2000F / 1090C	12.6	1.81

Abrasion Loss

per ASTM C 704
after 1500 F

7 cc

Coefficient of Thermal Expansion (reversible)

3.6 x 10⁻⁶ in/in F

6.4 x 10⁻⁶ mm/m C

Temperature per ASTM C113 / C865	Linear Change per ASTM C113 / C179	Cold MOR per ASTM C133 psi MPa	CCS per ASTM C133 psi MPa	Hot MOR per ASTM C583 psi MPa
230 F / 110 C	0.0%	2000 13.8	7000 48	
1500 F / 815 C	-0.2%	2500 17.2	8500 59	3900 26.9
2000 F / 1090C	-0.3%	2800 19.3	0	3600 24.8
2500 F / 1370 C	0.5%	3000 20.7	0	1000 6.9
3000 F / 1650 C	0.4%	3200 22.1	0	0.0

Other Data

Heat Up Guide

Schedule C or C-Linear

ASTM Class

C 401 Class F

Low Cement Castable

Notes:

All data are averaged results of ASTM tests (where applicable) on laboratory cast specimens. Reasonable variations in data can be expected. Data is not to be used for specification purposes. Product data is periodically updated to reflect product / raw material / process / testing changes. Please consult your Pilibrico representative to make sure you have the most current data.